

EEE4114F Class Test 2

May 2025

Duration: 1 hour

Total: 40 marks

Empl ID: _____

Instructions:

- This is a closed-book test.
- There are *four* questions totalling 40 marks. You must answer all of them.
- Questions should be answered in the space provided on the test script. If necessary, you may also write on the backs of the printed pages, but please indicate where you have done so.
- You must use a pen to complete the test – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all of your working clearly – your method and approach matter.
- Do not write your student number or name anywhere on this script; instead, provide only the 7-digit EMPLID found on your student card.
- A formula sheet is provided separately.
- You have 1 hour.

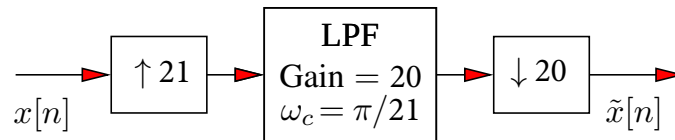
Question	Marks	Grade
1	10	
2	10	
3	10	
4	10	
Total:	40	

Question 1 [10 marks]

Consider a unity-gain, multirate system that converts an input signal with sampling frequency 10 kHz to an output signal with sampling frequency 10.5 kHz. Assume that the system uses a single ideal low-pass filter.

1. Draw a labelled block diagram for the system, specifying all relevant values.

Solution: Sketched as below. The input signal is first upsampled by a factor of $L = 21$, then low-pass filtered, and then downsampled by a factor of $M = 20$. This yields the desired overall sampling rate change of $21/20 = 10.5/10 = 1.05$.



Note that a single low-pass filter must be used, and with a gain of 20 (to get back to unity gain after the decimation process scales it by $1/M$) and a cut-off frequency of $\omega = \min(\pi/21, \pi/20) = \pi/21$ rad/sample.

2. Briefly explain the purpose of pre-filtering a signal before decimation.

Solution: The decimation process can easily result in spectral aliasing, so we apply a pre-filter (which may cost us some signal information) using a low-pass filter to avoid this.

3. What makes an ideal low-pass filter not physically realisable, and how could we attempt to address this?

Solution: An ideal low-pass filter is represented by a sinc-based impulse response in the time domain, which is infinitely long (not practical) and two-sided (non-causal). Thus we need to truncate and delay the signal such that it starts at/after $n = 0$, but this involves convolving the filter with a sinc function in frequency (and introduces a phase shift) – and hence it is no longer ideal.

4. Assume we have a bandlimited signal with bandwidth Ω_b . If we critically sampled this signal at the new sampling frequency of 10.5 kHz, find the value of Ω_b .

Solution: By Nyquist's criterion, we require that $\Omega_s \geq 2\Omega_b$, where $\Omega_s = 2\pi f_s = 2\pi(10500)$ (i.e., the sampling frequency in rad/sec). Thus, if the signal is critically sampled:

$$\begin{aligned}
 \Omega_s &= 2\Omega_b \\
 2\pi(10500) &= 2\Omega_b \\
 \therefore \Omega_b &= 10500\pi \text{ rad/sec}
 \end{aligned}$$

Question 2 [10 marks]

Consider the sequence $x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3]$, and let $X[k]$ be the 4-point DFT of $x[n]$.

1. Find the 4-point DFT $Y[k] = W_4^{2k} X[k]$.

Solution: The DFT of $x[n]$ is

$$X[k] = \sum_{n=0}^3 x[n] W_4^{kn} = W_4^{0k} + 2W_4^{1k} + 3W_4^{2k} + 4W_4^{3k}$$

As a result, we can find $Y[k]$ as:

$$\begin{aligned} Y[k] &= W_4^{2k} X[k] = W_4^{2k} + 2W_4^{3k} + 3W_4^{4k} + 4W_4^{5k} \\ &= 3W_4^{0k} + 4W_4^{1k} + W_4^{2k} + 2W_4^{3k} \end{aligned}$$

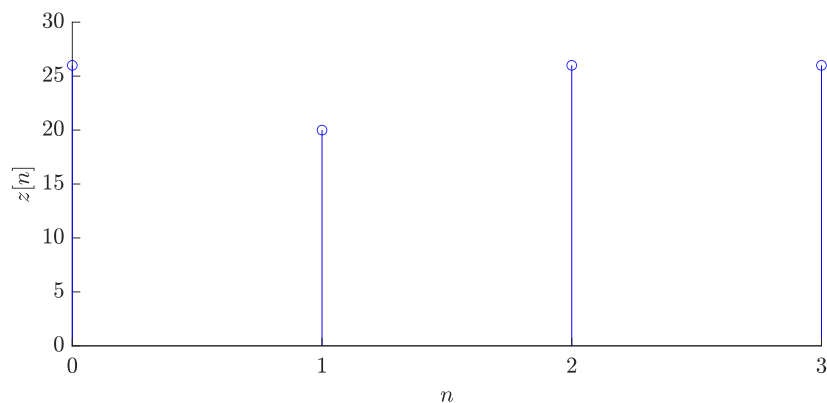
2. Find and sketch the 4-point circular convolution of $x[n]$ with $y[n]$.

Solution: 4-point circular convolution in the time domain corresponds to multiplication in the 4-point DFT domain. If $z[n] = x[n] \circledast y[n]$ is the circular convolution for $N = 4$, then in the DFT domain we have

$$\begin{aligned} Z[k] &= X[k]Y[k] = (W_4^{0k} + 2W_4^{1k} + 3W_4^{2k} + 4W_4^{3k})(3W_4^{0k} + 4W_4^{1k} + W_4^{2k} + 2W_4^{3k}) \\ &= 26W_4^{0k} + 20W_4^{1k} + 26W_4^{2k} + 26W_4^{3k} \end{aligned}$$

We can inverse transform this by inspection, and then sketch it as below:

$$z[n] = 26\delta[n] + 20\delta[n-1] + 26\delta[n-2] + 26\delta[n-3]$$



END OF PART A

Part B: Machine Learning

Question 3 [10 marks]

Suppose you have a dataset shown in Figure 1. The round circles represent training data, and black stars represent unseen test data.

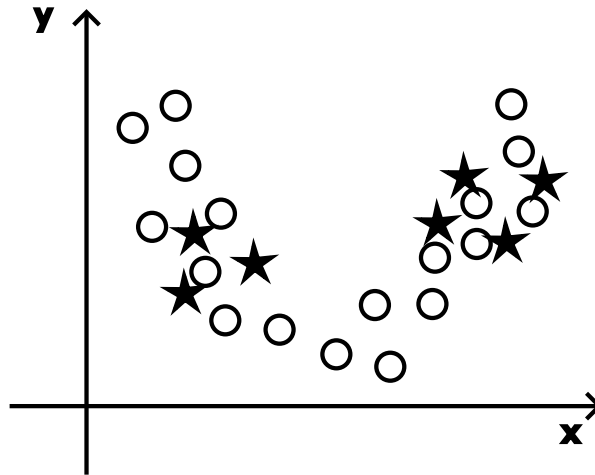


Figure 1: Plot of training and test data

You decide to train two different models on the dataset:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$k_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

Answer the following questions:

1. Which model is most likely to have high bias, but low variance? Why?

(2)

Solution: $h_{\theta}(x)$, as the data appears to be parabolic and requires features of x^2 . Without those features it would be unable to learn the task, and therefore result in high bias. There limited flexibility in the model due to a low capacity (only two parameters) and therefore would have low variance.

2. How could you improve the generalization performance of $k_{\theta}(x)$ without manually modifying the number of parameters or features? (2)

Solution: Add regularization to automatically constrain parameters and reduce model complexity.

3. Would it be possible for $h_{\theta}(x)$ to produce high r^2 scores on the test data? You may draw on Figure 1. to help answer the question if needed. (2)

Solution: The test data appears to be poorly sampled and not representative of the population. It would seem that a simple linear regression model could be trained that happens to pass through the majority of the test examples which would result in high r^2 scores.

4. What is a downside to random, repeated holdouts when resampling a small dataset? (2)

Solution: Information leak due to examples possibly appearing in validation more than once.

5. How would cross-validation improve the evaluation of this model? (2)

Solution: Cross-validation ensures that each example used during training also gets used for validation, but only once, as each validation set is independent.

Question 4 [10 marks]

Suppose you need to develop a system that estimates the distance of target from a sensor. You collect a data set, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) . x_i is the sensor's voltage, and y_i is the target variable, which, in this case, is distance to a target in meters.

$$\mathcal{D} = \{(1.1, 10), (1.2, 12), (1.1, 9), (0.9, 8)\}$$

You then decide to fit a regression model of the form, $h_\theta(x) = \theta_0 + \theta_1 x^2 + \theta_2 x^3$, to it using gradient descent optimization methods. The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

A single training iteration uses one training example and the model parameters are updated after **each** iteration.

The model's parameters after two training iterations are

$$\theta_0 = 0.5, \theta_1 = 1, \theta_2 = 2$$

The update rule for each parameter is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$.

Answer the following questions:

1. Write down the expression for the gradient of the objective function with respect to θ_2 . E.g. $\frac{\partial J}{\partial \theta_2}$. (1)

Solution:

$$\frac{\partial J}{\partial \theta_2} = ((\theta_0 + \theta_1 x_i^2 + \theta_2 x_i^3) - y_i) x_i^3$$

2. Calculate the gradient, $\frac{\partial J}{\partial \theta_2}$, after the third training iteration. (2)

Solution:

$$h_\theta(1.1) = 0.5 + 1(1.1^2) + 2(1.1^3) = 4.37$$

$$\frac{\partial J}{\partial \theta_2} = ((4.37 - 9)1.1^3) = -6.16$$

3. Calculate what the value for θ_2 would be after the **fourth** training iteration given that you have $\frac{\partial J}{\partial \theta_0} = -4, 63$, (5) and $\frac{\partial J}{\partial \theta_1} = -5.60$.

Solution: First, we need to know the parameters after the first update.

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 0.5 - 0.1(-4, 63) \\ &:= 0.96\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \eta \frac{\partial J}{\partial \theta_1} \\ &:= 1 - 0.1(-5.60) \\ &:= 1.56\end{aligned}$$

$$\begin{aligned}\theta_2 &:= \theta_2 - \eta \frac{\partial J}{\partial \theta_2} \\ &:= 2 - 0.1(-6.16) \\ &:= 2.62\end{aligned}$$

Then repeat the calculation of gradients, but using the new parameters and new data $h_\theta(x_i) = 0.96 + 1.56(x_i)^2 + 2.62(x_i)^3$

$$h_\theta(0.9) = 0.96 + 1.56(0.9)^2 + 2.62(0.9)^3 = 4.13$$

$$\frac{\partial J}{\partial \theta_2} = (4.13 - 8) * 0.9^3 = -2.82$$

Then the new value is

$$\begin{aligned}\theta_2 &:= \theta_2 - \eta \frac{\partial J}{\partial \theta_2} \\ &:= 2.62 - 0.1(-2.82) \\ &:= 2.90\end{aligned}$$

4. If you had a neural network with a single hidden layer which has two neurons, and one output neuron. Write (2)
down the expression for the bias value $\frac{\partial J}{\partial b_1^1}$ using chain rule

Solution: $\frac{\partial J}{\partial b_1^1} = \frac{\partial J}{\partial a_1^2} \frac{\partial a_1^2}{\partial z_1^2} \frac{\partial z_1^2}{\partial a_1^1} \frac{\partial a_1^1}{\partial z_1^1} \frac{\partial z_1^1}{\partial b_1^1}$

END OF PART B